

hyperparameters

for validation loss

FeedForward: Factor-of-4 rule of thumb (8/3 for GLVs) is ^{standard}

Head dim: Head dim * Num head = D model is standard

Aspect ratio: wide range of 'good' values (100-200) — but low to no validation

Regularization: you still 'regularize' LMs but its effects are primarily on optimize on dynamics

Stability tricks ^{badly behaved}

Softmax: can be ill-behaved due to exponentials/division by zero

① output softmax stability — the 'z-loss'

PaLM: pioneered this 'z loss' trick

② Attention softmax stability — the 'QK' norm'

logit & soft-capping: soft-capping the logits to some max value in Tanh

prevents logits from blowing-up, but also might have perf issue

Attention heads

GQA/MQA/MHQA: saving inference cost by reducing attention heads

$\Rightarrow$ **Goat/MQA**: $\rightarrow$ multiple queries, but just one dim for key/values
 Reducing attention head cost.

total arithmetic operations: bnd^2

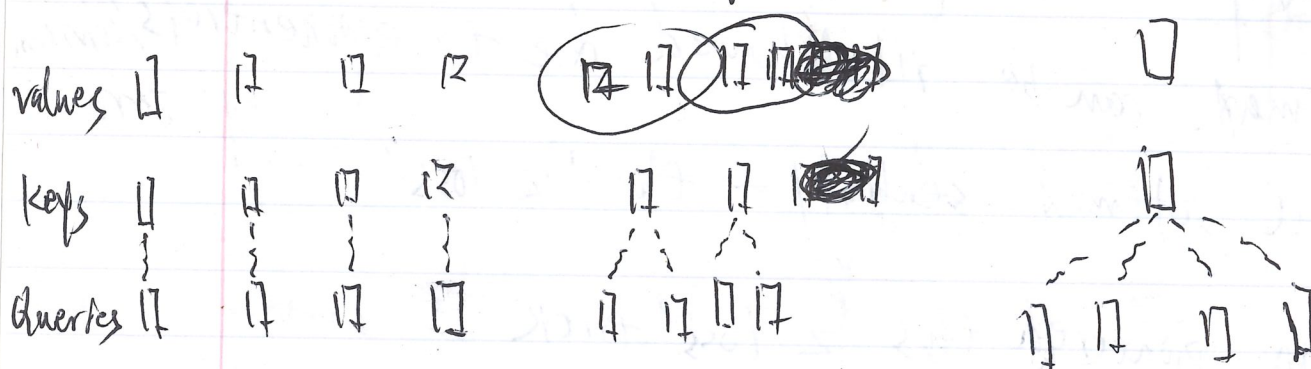
total memory accesses: $bnd + bnd^2 + d^2$

K-V cache

= traditional
 multi-head

Goat
 Grouped-query

MQA
 multi-query



control (K-Q ratio) and inference efficiency

$\Rightarrow$ **sparse/sliding window attention**

sparse/structure attention $\Rightarrow$ trade-off (expressiveness vs runtime)

sliding window attention $\Rightarrow$ dynamically compute attention

with sliding process